

Chapter 6. Device Attestation and Operational Credentials

This chapter describes the procedures and cryptographic credentials involved in establishing trust between entities.

The [Device Attestation](#) section provides mechanisms for Commissioners and Administrators to determine whether a Node is a genuine certified product before sharing sensitive information such as keys and other credentials. The Device Attestation feature relies on a [Device Attestation Certificate \(DAC\)](#) chain and on a [Certification Declaration \(CD\)](#).

The [Node Operational Credentials](#) section describes the credentials used by all Nodes to mutually authenticate each other during [Certificate-Authenticated Session Establishment](#), including the [Node Operational Certificate \(NOC\)](#) chain. These credentials form the basis of how Nodes are identified and take part in securing operational [unicast communication](#).

6.1. Certificate Common Conventions

This chapter makes use of digital certificates in several subsections. All certificates within this specification are based on X.509v3-compliant certificates as defined in [RFC 5280](#). The storage format of the certificates depends on application (e.g., DAC or NOC chain), but all certificates are directly compatible with X.509v3 DER representation after suitable loading or decompression.

In order to simplify further exposition, this subsection contains some common normative conventions that SHALL apply to all digital certificates described in this specification.

The following certificate formats are defined within this specification:

- Compressed [Node Operational credentials](#) certificate chain elements in [Matter Operational Certificate Encoding](#) or "Matter Certificate" format:
 - Node Operational Certificate ([NOC](#))
 - Intermediate CA Certificate ([ICAC](#))
 - Root CA Certificate ([RCAC](#))
- Device Attestation certificate chain elements in Standard X.509 DER format:
 - Device Attestation Certificate (DAC): see [Section 6.2.2.3, "Device Attestation Certificate \(DAC\)"](#)
 - Product Attestation Intermediate (PAI): see [Section 6.2.2.4, "Product Attestation Intermediate \(PAI\) Certificate"](#)
 - Product Attestation Authority (PAA): see [Section 6.2.2.5, "Product Attestation Authority \(PAA\) Certificate"](#)

6.1.1. Encoding of Matter-specific RDNs

In addition to the standard DN (Distinguished Names) attribute types that appear in certificate Subject and Issuer fields, there are Matter-specific DN attribute types under the [1.3.6.1.4.1.1.37244](#) pri-